# §6 正弦级数和余弦级数

- 奇函数和偶函数的傅里叶级数
- 函数展开成正弦级数或余弦级数

# 一、奇函数和偶函数的傅里叶级数

一般说来,一个函数的傅里叶级数既含有正弦项,又含有余弦项.但是,也有一些函数的傅里叶级数只含有正弦项或者只含有常数项和余弦项.

**定理** (1) 当周期为$2\pi$的奇函数$f(x)$展开成傅里叶级数时, 它的傅里叶系数为

$$a_n = 0 \qquad (n = 0,1,2,\cdots)$$

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx dx \qquad (n = 1,2,\cdots)$$

(2) 当周期为$2\pi$的偶函数$f(x)$展开成傅里叶级数时，它的傅里叶系数为

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx dx \quad (n = 0,1,2,\cdots)$$

$$b_n = 0 \qquad (n = 1,2,\cdots)$$

**证明** (1) 设$f(x)$是奇函数，

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \underbrace{f(x)\cos nx dx}_{\text{奇函数}} = 0 \qquad (n = 0,1,2,3,\cdots)$$

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$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \underbrace{f(x)\sin nx}_{\text{偶函数}} dx = \frac{2}{\pi}\int_{0}^{\pi} f(x)\sin nx dx$$

$$(n = 1,2,3,\cdots)$$

同理可证(2)　　　　　　　　　　定理证毕.

**定义** 如果$f(x)$为奇函数, 傅氏级数$\sum_{n=1}^{\infty} b_n \sin nx$

称为**正弦级数**.

如果$f(x)$为偶函数, 傅氏级数$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$

称为**余弦级数**.

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**例 1** 设 $f(x)$ 是周期为 $2\pi$ 的周期函数,它在 $[-\pi,\pi)$ 上的表达式为 $f(x)=x$,将 $f(x)$ 展开成傅氏级数.

**解** 所给函数满足狄利克雷充分条件.

在点 $x=(2k+1)\pi(k=0,\pm1,\pm2,\cdots)$ 处不连续,

收敛于 $\dfrac{f(\pi-0)+f(-\pi+0)}{2}=\dfrac{\pi+(-\pi)}{2}=0,$

在连续点 $x(x\neq(2k+1)\pi)$ 处收敛于 $f(x)$,

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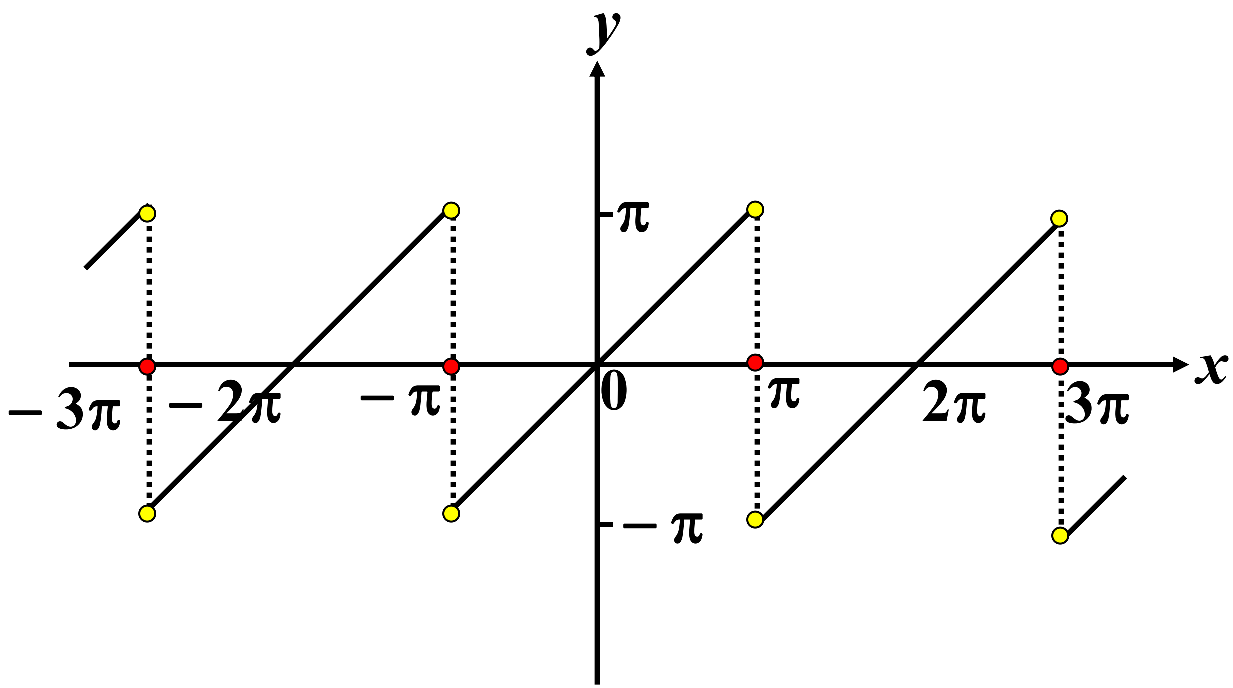
$\because x \neq (2k+1)\pi$时 $f(x)$是以$2\pi$为周期的奇函数,

和函数图象




$$\therefore a_n = 0, \qquad (n = 0,1,2,\cdots)$$

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$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx dx = \frac{2}{\pi}\int_0^{\pi} x\sin nx dx$$

$$= \frac{2}{\pi}\left[-\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}\right]_0^{\pi}$$

$$= -\frac{2}{n}\cos n\pi = \frac{2}{n}(-1)^{n+1}, \qquad (n = 1, 2, \cdots)$$

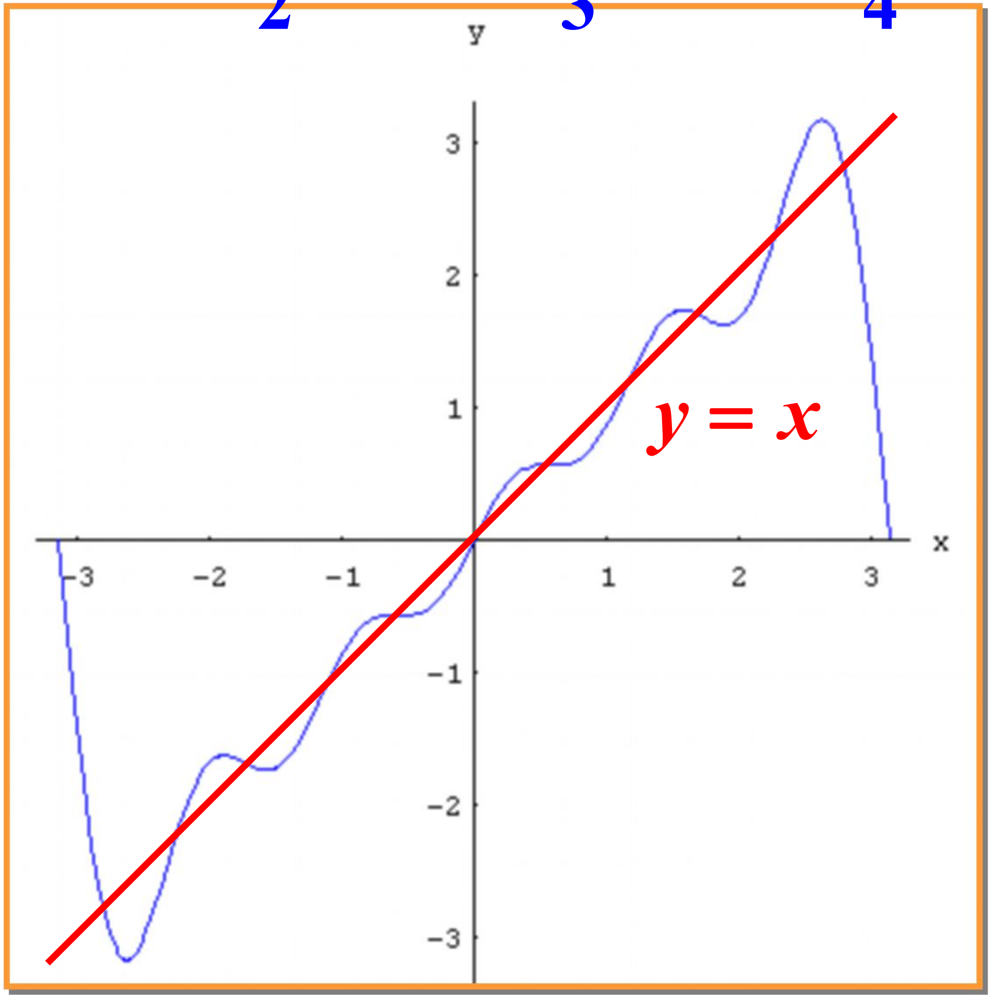
$$f(x) = 2(\sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - \cdots)$$

$$= 2\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin nx.$$

$$(-\infty < x < +\infty; x \neq \pm\pi, \pm 3\pi, \cdots)$$

$$y = 2(\sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - \frac{1}{4}\sin 4x + \frac{1}{5}\sin 5x)$$

观察两函数图形

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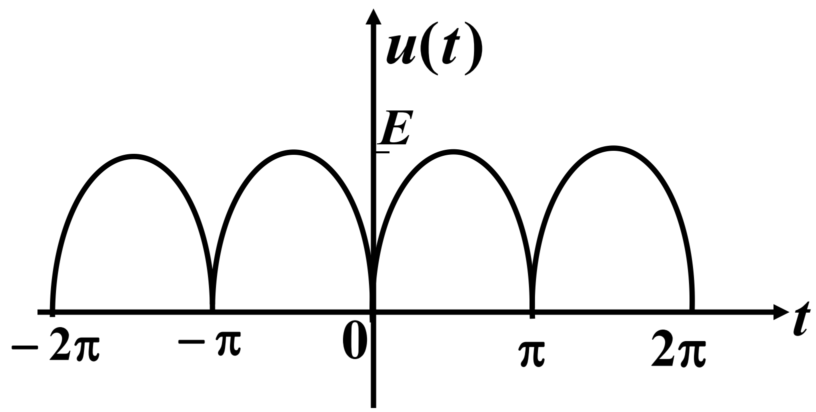
**例 2** 将周期函数$u(t)=|E\sin t|$展开成傅氏级数，其中$E$是正常数.

**解** 所给函数满足狄利克雷充分条件，在整个数轴上连续.

$\because u(t)$为偶函数，

$\therefore b_n=0,$

$(n=1,2,\cdots)$




$$a_0=\frac{2}{\pi}\int_0^{\pi}u(t)dt=\frac{2}{\pi}\int_0^{\pi}E\sin t\,dt=\frac{4E}{\pi},$$

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$$a_n = \frac{2}{\pi}\int_0^{\pi} u(t)\cos nt\,dt = \frac{2}{\pi}\int_0^{\pi} E\sin t\cos nt\,dt$$

$$= \frac{E}{\pi}\int_0^{\pi}[\sin(n+1)t - \sin(n-1)t]dt$$

$$= \frac{E}{\pi}\left[-\frac{\cos(n+1)t}{n+1} + \frac{\cos(n-1)t}{n-1}\right]_0^{\pi} \quad (n \neq 1)$$

$$= \begin{cases} -\dfrac{4E}{[(2k)^2-1]\pi}, & \text{当}n = 2k \\ 0, & \text{当}n = 2k+1 \end{cases} \quad (k = 1,2,\cdots)$$

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$$a_1 = \frac{2}{\pi}\int_0^{\pi} u(t)\cos t\,dt = \frac{2}{\pi}\int_0^{\pi} E\sin t\cos t\,dt = 0,$$

$$u(t) = \frac{4E}{\pi}\left(\frac{1}{2} - \frac{1}{3}\cos 2t - \frac{1}{15}\cos 4t - \frac{1}{35}\cos 6t - \cdots\right)$$

$$= \frac{2E}{\pi}\left[1 - 2\sum_{n=1}^{\infty}\frac{\cos 2nx}{4n^2-1}\right].$$

$$(-\infty < x < +\infty)$$

## 二、函数展开成正弦级数或余弦级数

### 非周期函数的周期性开拓

设$f(x)$定义在$[0,\pi]$上，延拓成以$2\pi$为周期的函数$F(x)$.

令 $F(x)=\begin{cases} f(x) & 0\le x\le \pi \\ g(x) & -\pi<x<0 \end{cases}$, 且$F(x+2\pi)=F(x)$,

则有如下两种情况 $\begin{cases}\text{奇延拓} \\ \text{偶延拓}\end{cases}$.

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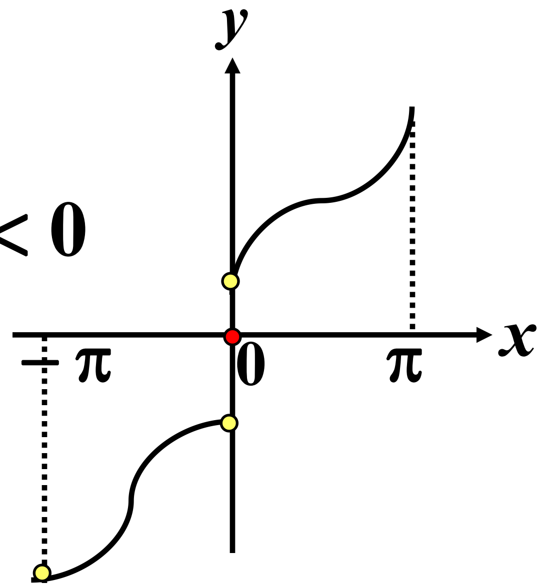
**奇延拓:** $g(x) = -f(-x)$

$$\text{则} F(x) = \begin{cases} f(x) & 0 < x \le \pi \\ 0 & x = 0 \\ -f(-x) & -\pi < x < 0 \end{cases}$$




$f(x)$的傅氏正弦级数

$$f(x) \leftrightarrow \sum_{n=1}^{\infty} b_n \sin nx \qquad (0 < x < \pi)$$

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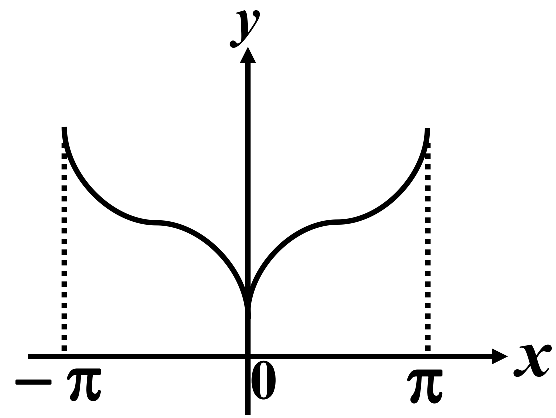
**偶延拓:** $g(x) = f(-x)$

则 $F(x) = \begin{cases} f(x) & 0 \le x \le \pi \\ f(-x) & -\pi < x < 0 \end{cases}$




$f(x)$的傅氏余弦级数

$$f(x) \leftrightarrow \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \qquad (0 \le x \le \pi)$$

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**例 3** 将函数 $f(x)=x+1\ (0\le x\le \pi)$ 分别展开成正弦级数和余弦级数.

**解** (1) 求正弦级数. 对 $f(x)$ 进行奇延拓,

$$F(x)=\begin{cases} f(x)=x+1 & 0<x\le\pi \\ 0 & x=0 \\ -f(-x)=x-1 & -\pi<x<0 \end{cases}$$

$$\text{则} b_n=\frac{2}{\pi}\int_0^{\pi}F(x)\sin nx\,dx=\frac{2}{\pi}\int_0^{\pi}f(x)\sin nx\,dx$$

例 3 将函数 $f(x)=x+1\ (0\le x\le \pi)$ 分别展开成正弦级数和余弦级数.

解 (1) 求正弦级数. 对 $f(x)$ 进行奇延拓,

$$b_n=\frac{2}{\pi}\int_0^{\pi}f(x)\sin nx dx=\frac{2}{\pi}\int_0^{\pi}(x+1)\sin nx dx$$

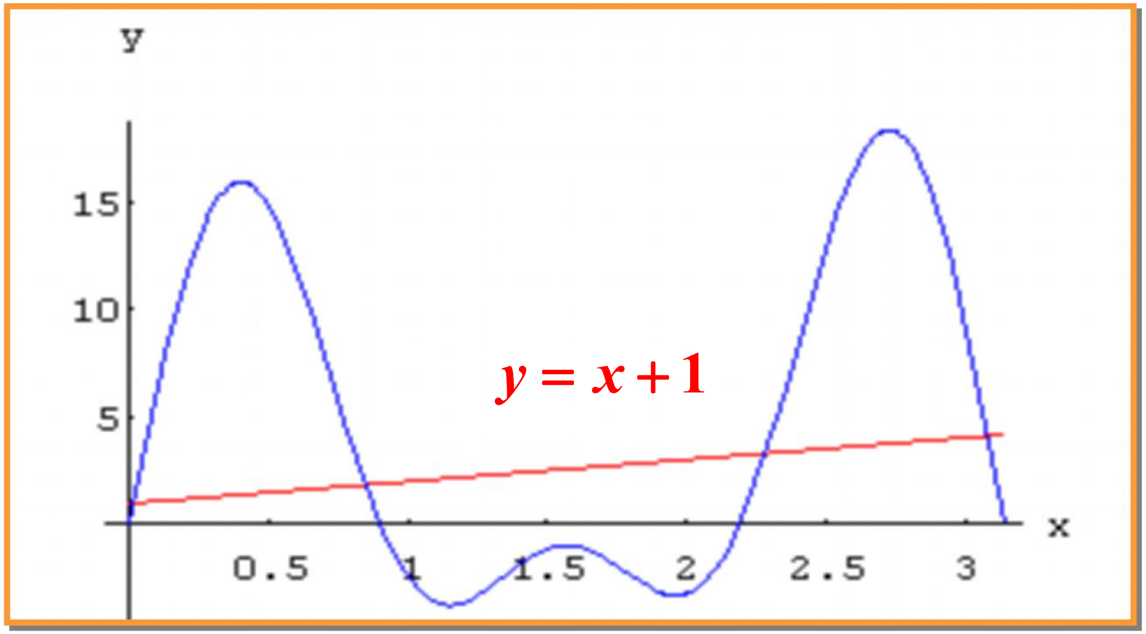
$$=\frac{2}{n\pi}(1-\pi\cos n\pi-\cos n\pi)$$

$$=\begin{cases}\dfrac{2}{\pi}\cdot\dfrac{\pi+2}{n} & \text{当} n=1,3,5,\cdots\\[2ex] -\dfrac{2}{n} & \text{当} n=2,4,6,\cdots\end{cases}$$

$$x+1=\frac{2}{\pi}[(\pi+2)\sin x-\frac{\pi}{2}\sin 2x+\frac{1}{3}(\pi+2)\sin 3x-\cdots]$$

$$(0<x<\pi)$$

$$x+1=\frac{2}{\pi}[(\pi+2)\sin x-\frac{\pi}{2}\sin 2x+\frac{1}{3}(\pi+2)\sin 3x-\frac{\pi}{4}\sin 4x+\frac{1}{5}(\pi+2)\sin 5x]$$

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(2) 求余弦级数.　　对$f(x)$进行偶延拓,

$$F(x)=\begin{cases} f(x)=x+1 & 0\le x\le\pi \\ f(-x)=-x+1 & -\pi<x<0 \end{cases}$$

则$a_0=\frac{2}{\pi}\int_0^{\pi}F(x)dx=\frac{2}{\pi}\int_0^{\pi}f(x)dx=\frac{2}{\pi}\int_0^{\pi}(x+1)dx=\pi+2,$

$$a_n=\frac{2}{\pi}\int_0^{\pi}F(x)\cos nxdx=\frac{2}{\pi}\int_0^{\pi}f(x)\cos nxdx$$

$$=\frac{2}{\pi}\int_0^{\pi}(x+1)\cos nxdx$$

(2) 求余弦级数. 对$f(x)$进行偶延拓,

$$a_0 = \frac{2}{\pi}\int_0^{\pi}(x+1)dx = \pi + 2,$$

$$a_n = \frac{2}{\pi}\int_0^{\pi}(x+1)\cos nx dx$$

$$= \frac{2}{n^2\pi}(\cos n\pi - 1) = \begin{cases} 0 & \text{当} n = 2,4,6,\cdots \\ -\dfrac{4}{n^2\pi} & \text{当} n = 1,3,5,\cdots \end{cases}$$

$$x + 1 = \frac{\pi}{2} + 1 - \frac{4}{\pi}(\cos x + \frac{1}{3^2}\cos 3x + \frac{1}{5^2}\cos 5x + \cdots]$$

$$(0 \le x \le \pi)$$

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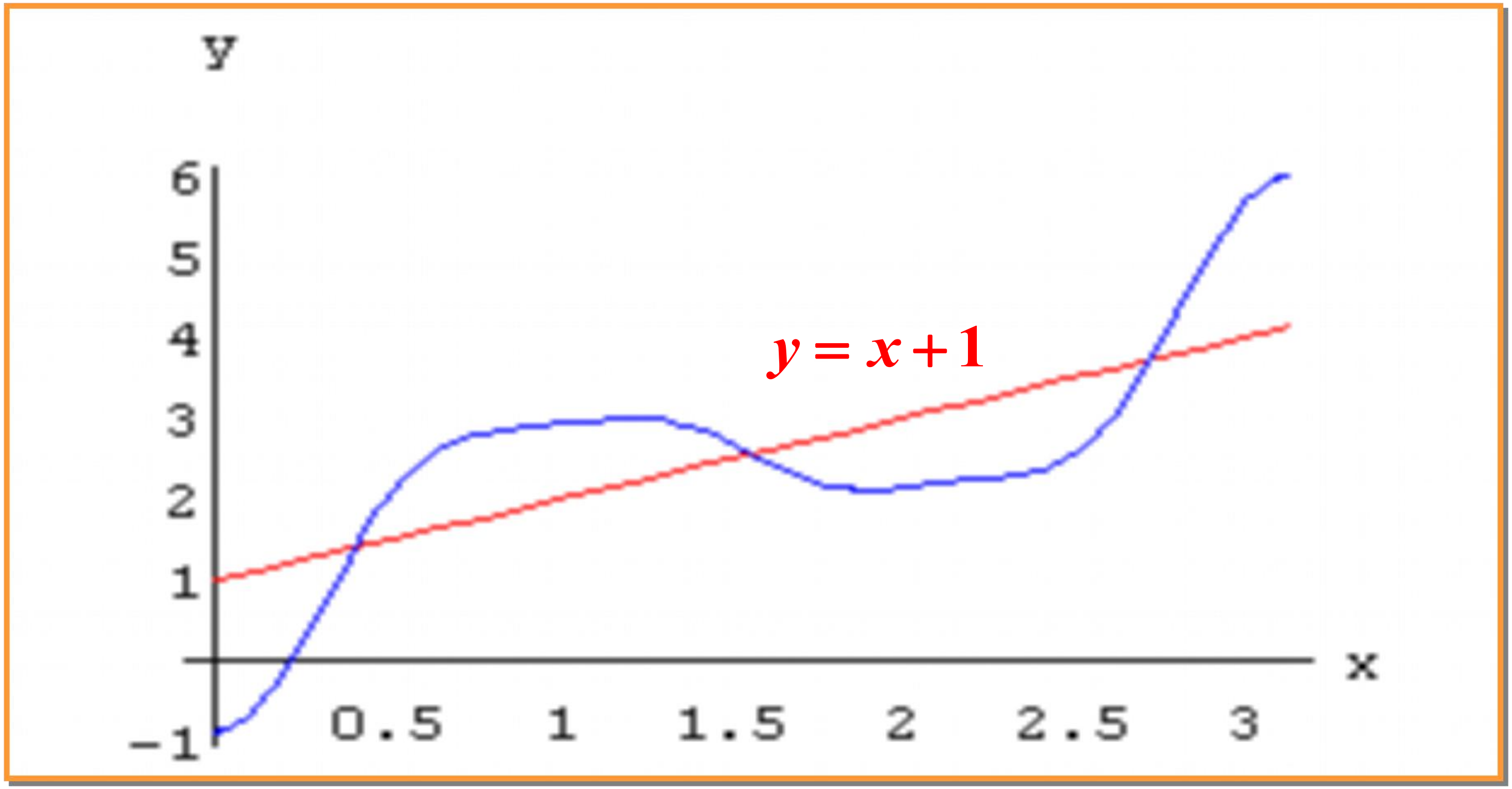
$$x+1=\frac{\pi}{2}+1-\frac{4}{\pi}\left(\cos x+\frac{1}{3^2}\cos 3x+\frac{1}{5^2}\cos 5x+\frac{1}{7^2}\cos 7x\right)$$

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例4 证明：当 $0 \le x \le \pi$ 时，$\sum_{n=1}^{\infty}\frac{\cos nx}{n^2} = \frac{x^2}{4} - \frac{\pi x}{2} + \frac{\pi^2}{6}$.

**解** 设 $f(x) = \frac{x^2}{4} - \frac{\pi x}{2}$,

将 $f(x)$ 在 $[0, \pi]$ 上展开成余弦级数：

$$a_0 = \frac{2}{\pi}\int_0^{\pi}\left(\frac{x^2}{4} - \frac{\pi x}{2}\right)dx = \frac{2}{\pi}\left(\frac{\pi^3}{12} - \frac{\pi^3}{4}\right) = -\frac{\pi^3}{3},$$

$$a_n = \frac{2}{\pi}\int_0^{\pi}\left(\frac{x^2}{4} - \frac{\pi x}{2}\right)\cos nx\,dx$$

$$=\frac{2}{n\pi}[(\frac{x^2}{4}-\frac{\pi x}{2})\sin nx\Big|_0^\pi-\int_0^\pi(\frac{x}{2}-\frac{\pi}{2})\sin nx dx]$$

$$=\frac{2}{n^2\pi}\int_0^\pi(\frac{x}{2}-\frac{\pi}{2})d\cos nx=\frac{2}{n^2\pi}\cdot\frac{\pi}{2}=\frac{1}{n^2}.$$

$$\therefore\quad \frac{x^2}{4}-\frac{\pi x}{2}=-\frac{\pi^2}{6}+\sum_{n=1}^{\infty}\frac{\cos nx}{n^2}.\qquad(0\le x\le\pi)$$

故 $$\sum_{n=1}^{\infty}\frac{\cos nx}{n^2}=\frac{x^2}{4}-\frac{\pi x}{2}+\frac{\pi^2}{6}.$$

# 小结

**1、基本内容:**

奇函数和偶函数的傅氏系数;正弦级数与余弦级数;非周期函数的周期性延拓;